

BTECH CSE · 7TH SEMESTER · K.P SINGH · SGT UNIVERSITY

Artificial Intelligence & Machine Learning

Lecture 1 · Introduction to Artificial Intelligence

How many AI systems have you already used **today**?



K.P Singh

Before We Define AI... a Story



"Beta, meri chashma bedroom mein kahin reh gayi hai. Zara dhoond ke laao."

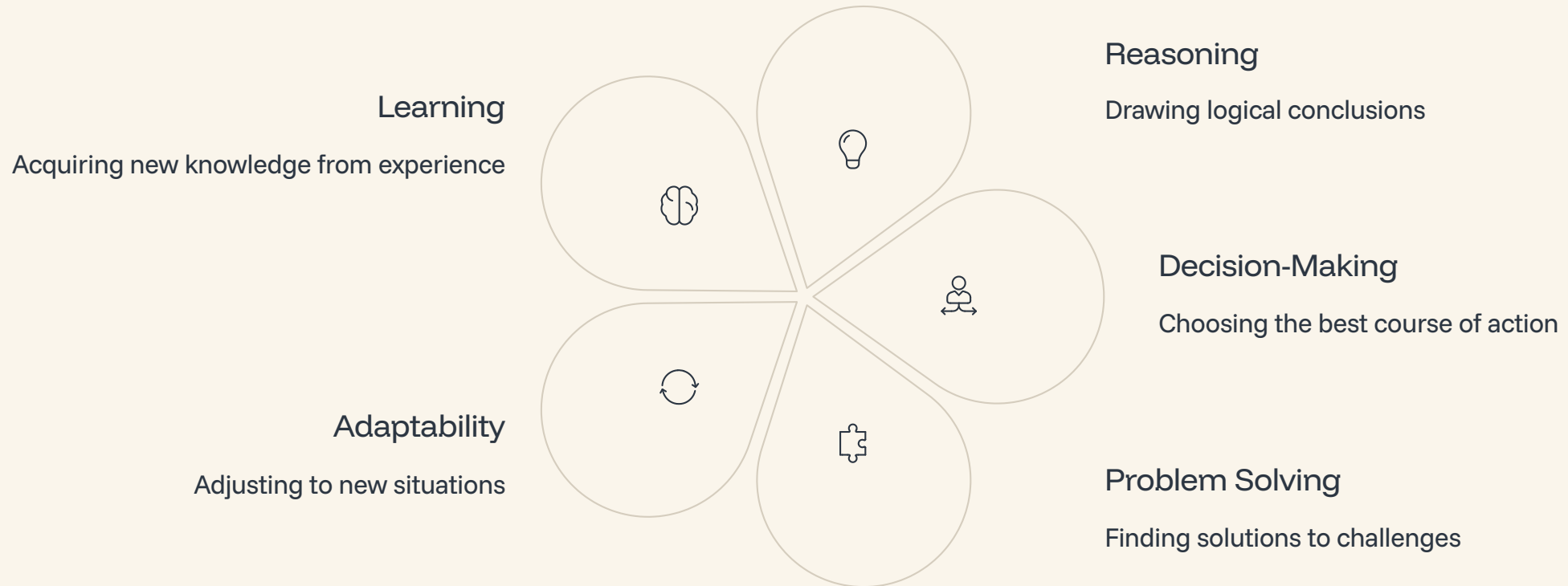
When your mother asks you to find her glasses, you don't search randomly. You check the bedside table first. Then the dressing table. Not the kitchen.

You didn't know where the glasses were — yet you searched **intelligently**.

That's the journey we start today.

What Makes Someone Intelligent?

Who is the most intelligent student in your class — and **why?** **Knowledge $\neq$ Intelligence.** It's about *how you use* what you know.."



Defining Artificial Intelligence

Artificial Intelligence is the science of making machines capable of performing tasks that normally require human intelligence. AI is not a robot. AI is not ChatGPT. **AI is an ability.**



Learn

Improve from experience and data



Reason

Draw conclusions from available information



Decide

Choose the best action



Solve Problems

Find solutions to complex challenges



Adapt

Adjust to new environments

AI Around Us — Spot the Intelligence

AI isn't about *displaying* information — it's about **making intelligent decisions from available information**.



Google Maps

Finds the **best route** in real time



Gmail

Filters **spam** automatically



Netflix

Recommends **movies** you'll love



Amazon

Suggests **products** based on behavior



Face Unlock

Recognizes your **face**

The Intelligent Agent

Anything that **perceives** its environment, **decides**, and **acts** to achieve a goal.

📄 **Perceive → Decide → Act** — the heart of every AI system.



Real-World Example: Swiggy Delivery Partner

- Receives order details
- Checks pickup location
- Observes live traffic
- Chooses optimal route
- Delivers on time

Quick Check — Agent or Not?

An Intelligent Agent doesn't have to look human. It just needs to **perceive, decide, and act**.



Calculator



Perceives input, but does it *decide*?



Google Maps



Perceives traffic, decides route, guides you



Chess AI



Perceives board, evaluates moves, plays



Thermostat



Senses temperature, decides, turns HVAC on/off



ChatGPT



Reads prompt, reasons, generates response



Robot Vacuum



Maps room, avoids obstacles, cleans

Exam-Ready Vocabulary

Artificial Intelligence

Science of making machines perform tasks requiring human intelligence

Intelligence

Ability to learn, reason, decide, solve problems, and adapt

Intelligent Agent

Entity that perceives its environment and takes actions to achieve goals

Perceive · Decide · Act

The three-step loop at the core of every AI agent

Likely Exam Questions

- 1 Define Artificial Intelligence with examples.
- 2 Define an Intelligent Agent.
- 3 Give real-life examples of Intelligent Agents.
- 4 Explain AI with real-world applications.

Today's Takeaways

AI $\neq$ Robots

AI is an **ability**, not a physical machine

AI = Perception + Action

Perceiving, deciding, learning, and acting intelligently

Agents Are Everywhere

From thermostats to ChatGPT — Intelligent Agents surround us

If an AI system must choose among *thousands* of possibilities — **how does it actually find the best one?**



Coming Next — Lecture 2

→ PROBLEM SOLVING BY SEARCH

STATE SPACE REPRESENTATION



The Question We Answer Next

Google Maps has **hundreds** of possible roads. A chess engine has **30** possible moves at each turn.

Does it guess? Or does it follow a **strategy**?

☐ → **Search**. The systematic process of exploring possibilities to find the best solution.

Next lecture: how AI agents navigate complex problem spaces using **search algorithms and state space representation**.